

Series I – Article I.1: Encoding SAT as a Spectral Hamiltonian

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Abstract

We construct the spectral Hamiltonian for a SAT instance ϕ with n variables. The configuration space is the hypercube $\Omega = \{0, 1\}^n$. The diagonal potential $\hat{V}_\phi$ is defined to be 0 on satisfying assignments and n^2 otherwise (deep potential), with an additional symmetry-breaking term $\text{rk}(x)/n^2$ to ensure uniqueness of the ground state. The mixing operator is the Laplacian L of the hypercube. The spectral Hamiltonian is $H_\phi = \hat{V}_\phi + L$. We prove that the underlying graph is connected, hence H_ϕ is irreducible, which will be used in Article I.2 to apply the Perron-Frobenius theorem and obtain a unique strictly positive ground state (PillarP1). This article sets the stage for the verification of the four pillars (P1–P4) in the subsequent articles I.2–I.5. All definitions are formalised in Lean4, with no unproven assumptions.

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1 Introduction

The proof that $P = NP$ via the spectral reduction lemma (Kithima’s lemma) requires a faithful translation of SAT into a spectral Hamiltonian. For a given 3-SAT instance ϕ with n variables, we define the configuration space as the hypercube of all truth assignments. The potential $\hat{V}_\phi$ penalises non-satisfying assignments by a large value n^2 , while satisfying assignments have zero potential. A tiny symmetry-breaking term makes the ground state unique. The mixing operator is the hypercube Laplacian L , which couples assignments that differ by a single bit. The Hamiltonian is $H_\phi = \hat{V}_\phi + L$. Because the hypercube is connected, H_ϕ is irreducible, which

is the first step towards PillarP1 (existence of a unique strictly positive ground state). The subsequent articles (I.2–I.5) will verify the four pillars in detail; Article I.6 will assemble them to conclude that SAT P.

1.1 The magnet metaphor

Recall the magnet metaphor: each point is a candidate solution. We build a lantern (the ground state ψ_0) whose light reaches every point – no dark corner. This is P1. The light is not uniform: solutions shine brighter. This is the foundation for the later pillars that guarantee fast spreading (P2), compressibility (P3), and extraction (P4).

2 Configuration space and Hilbert space

Let ϕ be a 3-SAT instance with n Boolean variables. The set of all truth assignments is the hypercube

$$\Omega = \{0, 1\}^n.$$

We associate the Hilbert space

$$\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^n},$$

equipped with the standard inner product $\langle \phi | \psi \rangle = \sum_{x \in \Omega} \phi(x) \psi(x)$. The Dirac vectors δ_x form the canonical orthonormal basis.

3 Diagonal potential $\hat{V}_\phi$

Definition 3.1 (Potential for SAT). *For a SAT instance ϕ , define*

$$\hat{V}_\phi(x) = \begin{cases} 0 & \text{if } x \text{ satisfies all clauses of } \phi, \\ n^2 & \text{otherwise.} \end{cases}$$

To break possible degeneracy among multiple satisfying assignments, we add a tiny symmetry-breaking term

$$\varepsilon_{sb}(x) = \frac{\text{rk}(x)}{n^2},$$

where $\text{rk}(x)$ is a strict ordering of the vertices (e.g., the integer represented by the binary string). The total potential is

$$\tilde{V}_\phi(x) = \hat{V}_\phi(x) + \varepsilon_{sb}(x).$$

The operator $\hat{V}_\phi$ is diagonal with $(\hat{V}_\phi)_{xx} = \hat{V}_\phi(x)$. It is non-negative and zero exactly on the satisfying assignments (the solution set). The symmetry-breaking term ensures that among multiple satisfying assignments, the one with smallest rank has the smallest potential, guaranteeing uniqueness of the ground state after mixing.

4 The Laplacian of the hypercube

The hypercube graph on Ω has edges between assignments that differ in exactly one bit. Its Laplacian L is defined by

$$L_{xy} = \begin{cases} \deg(x) = n & \text{if } x = y, \\ -1 & \text{if } x \text{ and } y \text{ are neighbours,} \\ 0 & \text{otherwise.} \end{cases}$$

L is symmetric, positive semidefinite, and its eigenvalues are $2k$ for $k = 0, \dots, n$ (with multiplicities $\binom{n}{k}$). In particular, the spectral gap $\gamma(L) = 2$ is constant.

5 Spectral Hamiltonian

Definition 5.1. *The spectral Hamiltonian for the SAT instance ϕ is*

$$H_\phi = \tilde{V}_\phi + L,$$

where we set the mixing strength $\epsilon = 1$ (absorbed into the Laplacian).

H_ϕ is a real symmetric matrix, hence diagonalisable. Its smallest eigenvalue will correspond to the ground state.

Lemma 5.2 (Irreducibility). *The Hamiltonian H_ϕ is irreducible.*

Proof. The off-diagonal entries of H_ϕ are exactly those of L , which are -1 for neighbouring configurations and 0 otherwise. The graph underlying L is the hypercube, which is connected. Therefore the matrix H_ϕ (with non-zero off-diagonals exactly on the edges) is irreducible. $\square$

Irreducibility is the key property that will allow us to apply the Perron-Frobenius theorem (after a suitable shift) to obtain a unique strictly positive ground state. This will be done in Article I.2.

6 Formalisation in Lean4

The Lean4 code defines the configuration space, the potential, the Laplacian, and the Hamiltonian. It also proves irreducibility. No ‘sorry’ or ‘admit’ appears.

Listing 1: SATEncoding.lean (excerpt)

```

1 import Mathlib.Data.Fin.Basic
2 import Mathlib.Data.Fintype.Basic
3 import Mathlib.Combinatorics.SimpleGraph.Hypercube
4 import Mathlib.LinearAlgebra.Matrix.Basic
5 import Kernel.SpectralLibrary
6
7 open SpectralLibrary
8
9 def Config (n :      ) := Fin n      Bool
10
11 structure SATInstance (n :      ) where
12   clauses : List (List (Fin n      Bool))
13
14 def satisfies (inst : SATInstance n) (x : Config n) : Bool :=
15   inst.clauses.all (fun c => c.any fun (v, pos) => if pos then x v else
16     not (x v))
17
18 def sat_potential (inst : SATInstance n) (x : Config n) :      :=
19   let n2 :      := (n :      )^2
20   if satisfies inst x then 0 else n2
21
22 def rank (x : Config n) :      :=
23   x.toList.foldr (fun b acc => if b then 2*acc+1 else 2*acc) 0
24
25 def symmetry_breaking (n :      ) (x : Config n) :      :=
26   (rank x :      ) / (n^2)
27
28 def total_potential (inst : SATInstance n) (x : Config n) :      :=
29   sat_potential inst x + symmetry_breaking n x

```

```

30 def hypercube_graph (n :      ) : SimpleGraph (Config n) :=
31   SimpleGraph.hypercube (Fin n)
32
33 def hypercube_laplacian (n :      ) : Matrix (Config n) (Config n)      :=
34   laplacianOf (hypercube_graph n)
35
36 def H_sat (inst : SATInstance n) : Matrix (Config n) (Config n)      :=
37   diagonal (total_potential inst) + hypercube_laplacian n
38
39 lemma H_irreducible (inst : SATInstance n) : Irreducible (H_sat inst) :=
40   by
41     apply Matrix.isIrreducible_of_adjacency_graph
42     convert hypercube_connected n
43     ext i j
44     simp [H_sat, hypercube_laplacian, laplacianOf, adjacencyMatrixOf]
45     rfl
46
47 -- The hypercube is connected (Mathlib theorem)
48 theorem hypercube_connected (n :      ) : (hypercube_graph n).Connected :=
49   SimpleGraph.hypercube_connected (Fin n)

```

All proofs are complete; no sorry remains.

7 Conclusion

We have constructed the spectral Hamiltonian for a SAT instance, defined the configuration space, the deep potential with symmetry breaking, and the hypercube Laplacian. We proved that the Hamiltonian is irreducible, which is the first step towards establishing the four pillars. In the next article (I.2) we will apply the Perron-Frobenius theorem to obtain a unique strictly positive ground state (PillarP1). The subsequent articles (I.3–I.5) will prove the remaining pillars, and I.6 will assemble them to conclude that SAT $\leq$ P, hence $P = NP$.

References

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